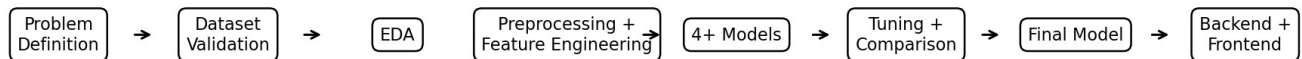


Predictive Maintenance Mini Project

Full Project Proposal, Dataset Analysis and Implementation Plan

Planned End-to-End FDM Workflow



Recommended project question: Can industrial equipment failure risk be predicted using machine operating conditions, sensor measurements, usage information and machine characteristics?

Dataset: UCI AI4I 2020 Predictive Maintenance Dataset

Prepared for a 4-member group | Binary Classification | 4+ Model Comparison

Document Overview

- Problem definition and real-world justification
- Prediction objective, stakeholders, inputs and outputs
- Dataset source, structure, variables and limitations
- Initial data-quality observations from the uploaded CSV
- EDA and preprocessing plan
- Feature-engineering plan
- Four-model development and comparison plan
- Evaluation metrics, class-imbalance strategy and leakage prevention
- Hyperparameter tuning and final-model selection
- Four-member contribution plan and viva preparation
- Backend/frontend architecture and demonstration plan
- Proposal-ready summary and references

1. Real-World Problem

Industrial equipment can fail unexpectedly during operation. Unplanned failures can cause production downtime, repair expense, delayed output and operational disruption. Reactive maintenance acts only after a breakdown, while fixed preventive-maintenance schedules may service healthy equipment unnecessarily. A predictive-maintenance system uses operating measurements to identify equipment that shows conditions associated with elevated failure risk.

Focused problem statement: Maintenance teams need an evidence-based way to identify machines that are at higher risk of failure before a breakdown, using measurements already available from the machine or process.

2. Prediction Objective and Scope

Item	Project definition
Prediction objective	Predict whether the current machine/process observation is associated with machine failure.
Target variable	Machine failure (0 = no failure, 1 = failure).
ML task	Binary classification.
Primary users	Maintenance engineers, plant supervisors, production/operations managers.
Decision supported	Prioritize equipment for inspection, monitoring or preventive maintenance.
System inputs	Machine type, temperatures, rotational speed,

	torque and tool wear.
System output	Failure/no-failure prediction; probability or risk score if supported by the selected model.
Project boundary	One focused failure-risk prediction problem. The project does not attempt to predict remaining useful life, fault diagnosis and maintenance scheduling simultaneously.

3. Why Predictive Maintenance Fits the Assignment

The assignment requires a complete data-mining workflow: problem understanding, validated dataset selection, EDA, preprocessing, feature engineering, at least four suitable machine-learning algorithms, optimization, final-model selection and an end-to-end prediction system. Predictive maintenance naturally supports all of these stages.

- Clear target variable and business/engineering purpose.
- Mixed categorical and numerical features.
- Class imbalance that requires thoughtful evaluation rather than accuracy alone.
- Potential outliers and nonlinear relationships in sensor/operating variables.
- Strong feature-engineering opportunities based on physical relationships.
- Meaningful model comparison across linear, tree-based and margin-based algorithms.
- Straightforward frontend demonstration for maintenance staff.
- Strong viva topics: leakage, scaling, imbalance, false-negative cost, feature importance and model interpretability.

4. Selected Dataset

Recommended dataset: AI4I 2020 Predictive Maintenance Dataset from the UCI Machine Learning Repository. UCI describes it as a synthetic dataset designed to reflect predictive-maintenance data encountered in industry. It contains 10,000 records and has no missing values according to the repository.

Dataset property	Details
Name	AI4I 2020 Predictive Maintenance Dataset
Repository	UCI Machine Learning Repository
Records	10,000
CSV columns	14
Main predictive features	6: Type, air temperature, process temperature, rotational speed, torque, tool wear
Primary target	Machine failure

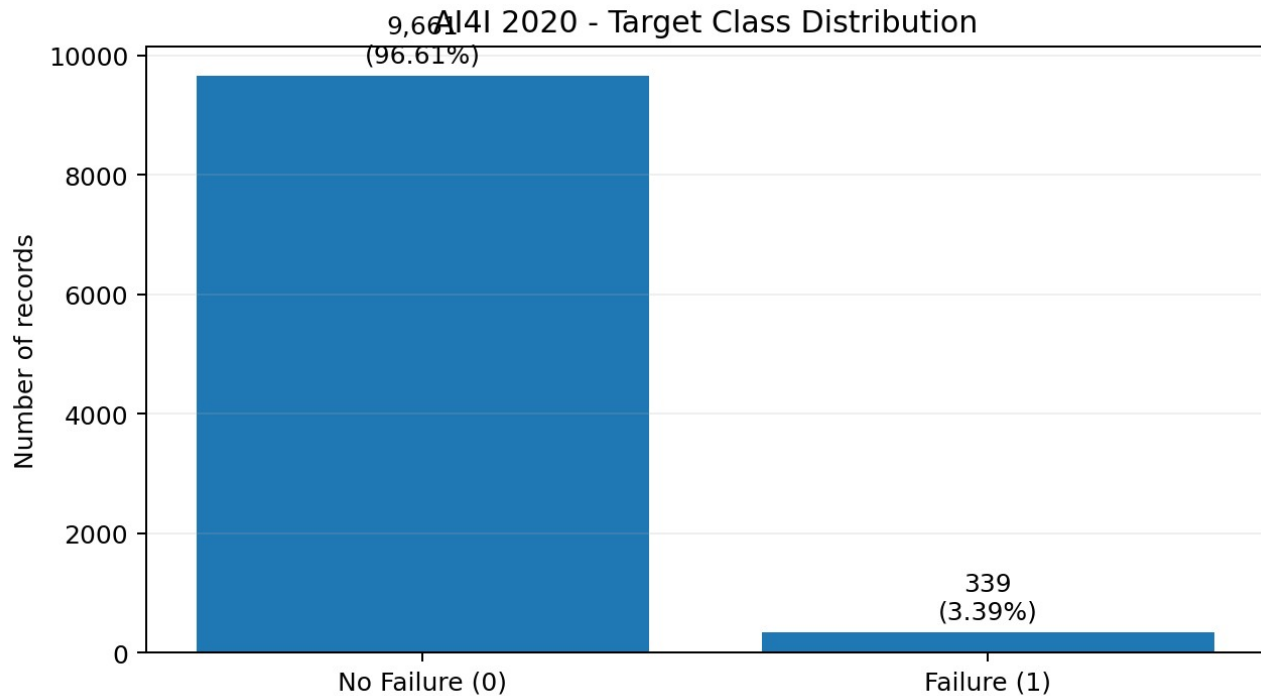
Missing values in uploaded CSV	0
Exact duplicate rows in uploaded CSV	0
License	Creative Commons Attribution 4.0 International (CC BY 4.0)
DOI	10.24432/C5HS5C
Important limitation	Synthetic rather than measurements from a single real industrial plant.

5. Dataset Variables and Modelling Role

Column	Role	Plan	Reason
UDI	Identifier	Exclude	Unique row identifier; not a physical predictor.
Product ID	Identifier	Usually exclude	Contains variant/serial information; Type already captures the quality category.
Type	Categorical feature	Use	Product quality variant: L, M or H.
Air temperature [K]	Continuous feature	Use	Operating/ environmental temperature.
Process temperature [K]	Continuous feature	Use	Process temperature.
Rotational speed [rpm]	Numeric feature	Use	Machine operating speed.
Torque [Nm]	Continuous feature	Use	Mechanical load indicator.
Tool wear [min]	Numeric feature	Use	Accumulated tool-wear time.
Machine failure	Target	Target	1 when the process fails; otherwise 0.
TWF	Failure-mode flag	Exclude from main model	Tool Wear Failure; can leak the overall failure outcome.
HDF	Failure-mode flag	Exclude from main model	Heat Dissipation Failure; can leak the

			overall failure outcome.
PWF	Failure-mode flag	Exclude from main model	Power Failure; can leak the overall failure outcome.
OSF	Failure-mode flag	Exclude from main model	Overstrain Failure; can leak the overall failure outcome.
RNF	Failure-mode flag	Exclude from main model	Random Failure; can leak the overall failure outcome.

6. Initial Data-Quality Findings from the Uploaded CSV



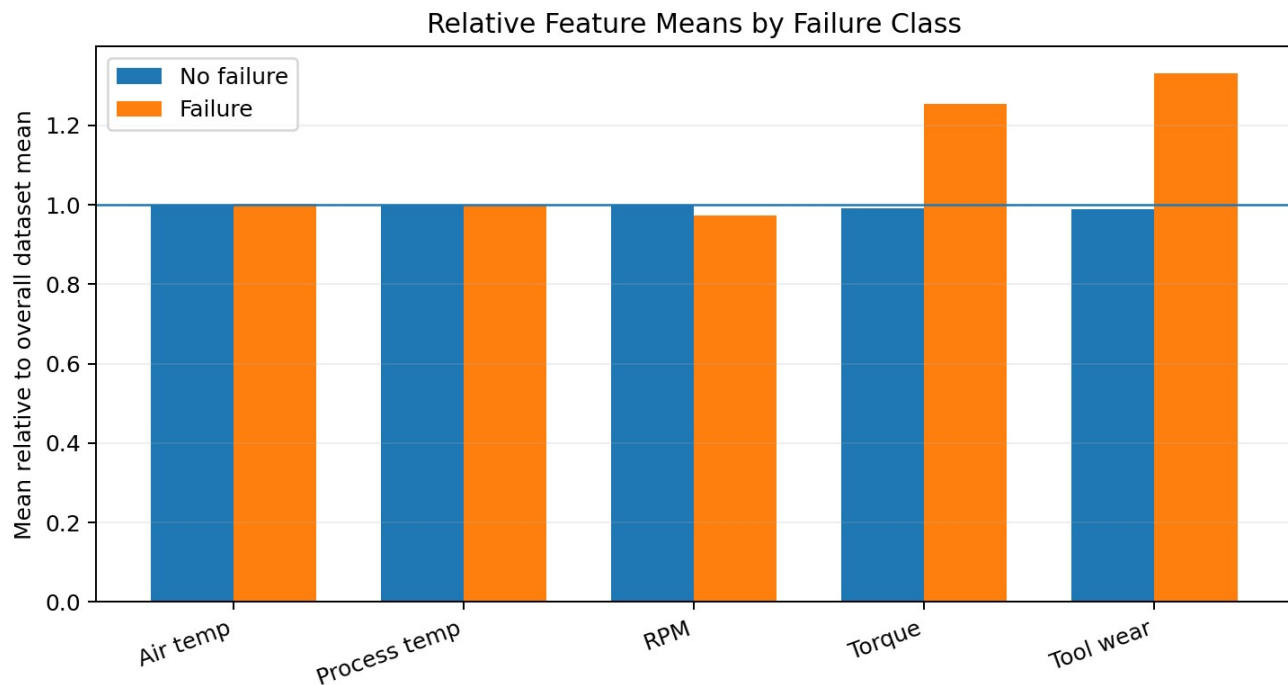
The uploaded CSV has 10,000 rows and 14 columns. No missing values or exact duplicate rows were found. The target is strongly imbalanced: 9,661 records (96.61%) are no-failure observations and 339 records (3.39%) are failure observations.

Why this matters: A trivial classifier that always predicts 'no failure' would achieve about 96.61% accuracy. Therefore, accuracy alone would be misleading. Recall, precision, F1-score and class-aware validation are essential.

7. Numerical Feature Snapshot

Feature	Mean	Std. dev.	Min	Median	Max
Air temperature [K]	300.00	2.00	295.30	300.10	304.50
Process temperature [K]	310.01	1.48	305.70	310.10	313.80
Rotational speed [rpm]	1538.78	179.28	1168.00	1503.00	2886.00
Torque [Nm]	39.99	9.97	3.80	40.10	76.60

Tool wear [min]	107.95	63.65	0.00	108.00	253.00
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The chart above is descriptive only. It compares class-specific means after dividing by the overall feature mean, so variables with different units can be viewed together. It should not be treated as proof of causality or as a final feature-selection decision.

8. Failure-Mode Flags and Leakage Risk

Failure-mode column	Positive records	Main-model treatment
TWF	46	Exclude
HDF	115	Exclude
PWF	95	Exclude
OSF	98	Exclude
RNF	19	Exclude

These columns identify specific failure modes and are logically downstream of the same event represented by the 'Machine failure' target. Using them as predictors could allow the model to infer the answer from information that would not be available as a genuine pre-failure input. They may be explored during EDA, but the primary predictive model should be trained without them.

9. Planned Exploratory Data Analysis (After Lecturer Validation)

- Check data types, unique values and identifier behaviour.

- Confirm missing values, duplicates and invalid ranges.
- Visualize class distribution and quantify imbalance.
- Plot numerical distributions overall and by target class.
- Investigate outliers in RPM, torque, temperature and tool wear.
- Compare failure rates across product Type categories.
- Examine pairwise relationships and correlations without interpreting correlation as causation.
- Study interactions such as torque vs rotational speed, tool wear vs torque and temperature difference vs failure.
- Check whether engineered variables improve separation between failure and non-failure observations.
- Document all findings that affect later preprocessing and model decisions.

10. Planned Preprocessing Pipeline

Step	Planned treatment	Reason
Identifiers	Remove UDI; normally remove Product ID.	Prevent models from learning row/serial identifiers.
Failure-mode columns	Remove TWF, HDF, PWF, OSF, RNF from predictors.	Prevent target leakage.
Categorical encoding	One-hot encode Type.	Convert categorical values to model-ready features.
Missing values	Verify; no missing values currently observed.	Pipeline should still handle unexpected missing input at system level.
Outliers	Investigate first; do not automatically delete.	Extreme operating values may be genuine and important for failure prediction.
Scaling	Standardize numerical features for Logistic Regression/SVM/KNN if used.	Distance/margin-based models are scale-sensitive.
Train/test split	Use stratification by Machine failure.	Preserve the rare failure proportion.
Cross-validation	Use stratified folds for model selection/tuning.	More reliable estimates with an imbalanced target.
Imbalance handling	Compare class weights and, if justified, resampling inside training folds only.	Avoid contamination of validation/test data.

11. Feature Engineering Opportunities

Feature engineering should be justified by domain logic and validated experimentally. New features should be created inside the training pipeline where appropriate to avoid leakage.

Candidate feature	Possible formula / idea	Reason to investigate
Temperature difference	Process temperature - Air temperature	May represent heat-transfer / process thermal gap.
Mechanical power proxy	Torque x rotational speed	Combines load and speed into a physically meaningful interaction proxy.
Wear-load interaction	Tool wear x Torque	A worn tool under high load may behave differently from either factor alone.
Wear-speed interaction	Tool wear x Rotational speed	Tests whether wear effect changes with operating speed.
Temperature-load interaction	Temperature difference x Torque	Tests joint thermal/mechanical stress.
Binned operating zones	Low/normal/high ranges after EDA	Could support interpretation, but should only be used if justified by data/domain evidence.
Important: Do not invent arbitrary features simply to increase model accuracy. Keep engineered variables only when they have a defensible rationale and improve validation performance without leakage.		

12. Four-Model Development Plan

All models solve the same binary classification problem using the same validated dataset and target. The assignment requires at least four suitable algorithms; the following set gives a useful range of modelling assumptions.

Model	Why include it	Key discussion points
Logistic Regression	Interpretable linear baseline.	Scaling, coefficients, class weights, threshold, linear decision boundary.
Decision Tree	Highly interpretable nonlinear model.	Gini/entropy, max depth, overfitting, pruning, feature splits.
Random Forest	Robust ensemble of trees.	Bagging, feature importance, n_estimators, max_depth, class_weight.
Support Vector Machine	Strong margin-based nonlinear/linear classifier.	Scaling, kernel choice, C, gamma, class weights.
Optional: Gradient Boosting	Strong additional nonlinear benchmark.	Learning rate, estimators, depth, bias-variance trade-off.

13. Evaluation Strategy

Because failures form only a small minority of the observations, model evaluation must focus on the minority class rather than accuracy alone.

Metric	Why it matters
Confusion matrix	Shows true/false positives and true/false negatives directly.
Recall (failure class)	Measures the proportion of actual failures detected. Important when missed failures are costly.
Precision (failure class)	Measures how many predicted failures are actually failures; helps control unnecessary alarms.
F1-score	Balances precision and recall.
ROC-AUC	Measures ranking ability across thresholds; useful but can look optimistic with severe imbalance.
PR-AUC	Often more informative for rare positive classes

	because it focuses on precision-recall trade-offs.
Accuracy	Can still be reported, but must not be the main selection criterion in this dataset.

14. False Positives vs False Negatives

Case	Meaning in this project	Operational implication
False positive	Model predicts failure risk but the machine would not fail.	May trigger unnecessary inspection or maintenance.
False negative	Model predicts no failure while a failure actually occurs.	Potentially leaves a high-risk machine uninspected; often more serious in predictive maintenance.

The final decision threshold should therefore be selected using the intended operating trade-off, not simply the default 0.50 probability threshold. For this academic project, the group should explain the trade-off rather than claim that one threshold is universally correct.

15. Hyperparameter Tuning Plan

Model	Example parameters to tune
Logistic Regression	C, penalty/solver (compatible combinations), class_weight.
Decision Tree	max_depth, min_samples_split, min_samples_leaf, criterion, class_weight.
Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features, class_weight.
SVM	C, kernel, gamma, class_weight.
Gradient Boosting (optional)	n_estimators, learning_rate, max_depth / tree complexity, subsampling parameters as applicable.

Use an appropriate search strategy such as GridSearchCV or RandomizedSearchCV with stratified cross-validation. Any oversampling/resampling must be applied only inside the training folds, ideally through a pipeline.

16. Final-Model Selection

- Compare untuned baselines using the same data split and metrics.
- Tune suitable models using stratified cross-validation.
- Compare tuned models against their baselines.

- Consider minority-class recall/precision/F1 or PR-AUC, not accuracy alone.
- Check stability across folds and inspect confusion matrices.
- Prefer a model whose improvement is consistent and whose complexity can be justified.
- Record why the final model is selected; do not select it solely because one metric is marginally highest.

17. Suggested Four-Member Work Allocation

Member	Primary technical ownership	Viva-ready responsibility
Member 1	EDA + Logistic Regression	Class distribution, distributions/relationships, baseline model, imbalance implications.
Member 2	Preprocessing + Decision Tree	Encoding, leakage prevention, outlier policy, tree split logic and overfitting.
Member 3	Feature Engineering + Random Forest	Engineered variables, feature importance, ensemble reasoning and tuning.
Member 4	Scaling/Optimization + SVM	Standardization, SVM kernel/C/gamma, cross-validation and tuning strategy.
All members	Final comparison + system integration	Every member must understand the whole workflow and be able to justify final decisions.

18. Viva Discussion Topics

- Why is this a classification problem?
- Why did you choose this dataset rather than simply choosing an easy dataset?
- Why are UDI and Product ID not useful predictors?
- Why must TWF/HDF/PWF/OSF/RNF be excluded from the main input set?
- Why is 96%+ accuracy potentially meaningless here?
- Which metric would you prioritize and why?
- Why does SVM require scaling while Decision Trees generally do not?
- Why should oversampling be performed only on training data/folds?
- How did you investigate outliers, and why might deleting them be harmful?
- What engineered features did you test and what evidence justified keeping or rejecting them?
- Why did the selected final model perform differently from the other models?
- What are the limitations of using a synthetic dataset?

19. Proposed End-to-End System

After the final model and preprocessing pipeline are selected, they can be packaged into a backend service. The final frontend should be a simple maintenance decision-support interface rather than a complex industrial control system.

Layer	Planned function
Frontend	Form/dashboard for Type, air temperature, process temperature, RPM, torque and tool wear.
Input validation	Check required fields, numeric ranges and permitted Type values.
Backend API	Receive input, run the saved preprocessing pipeline, call the trained final model and return result.
Model artifact	Serialized final model plus exactly the preprocessing used during training.
Output	Failure/no-failure prediction and, where available, probability/risk score.
Logging/testing	Test normal, boundary, missing and invalid input cases.

20. Example User Interaction

Example input	Example output concept
Type: M Air temperature: 300.2 K Process temperature: 310.4 K Rotational speed: 1450 rpm Torque: 52.0 Nm Tool wear: 185 min	Predicted status: AT RISK / NO FAILURE RISK Estimated probability: shown if supported by final model Note: Academic decision-support output; not a certified industrial safety system.

21. Risks and Limitations

- The AI4I dataset is synthetic; performance on this dataset does not prove real-factory deployment performance.
- The target is highly imbalanced, so model evaluation can be unstable if validation is poorly designed.
- Failure-mode labels create an obvious leakage risk if included as predictors.
- The dataset is relatively small for rare-event modelling: only 339 positive target records are present.

- Sensor relationships in a synthetic dataset may be cleaner or more structured than real industrial sensor streams.
- The final application should be described as a decision-support prototype, not a replacement for engineering inspection or safety procedures.

22. Dataset Proposal - Submission-Ready Summary

Project title: Machine Learning-Based Predictive Maintenance for Industrial Equipment Failure Risk Detection.

Problem: Unexpected equipment failures can cause unplanned downtime, repair cost and operational disruption. Maintenance teams need a method to identify machines showing operating conditions associated with elevated failure risk before breakdown.

Prediction objective: Predict whether an industrial machine/process observation is associated with machine failure.

Dataset: AI4I 2020 Predictive Maintenance Dataset, UCI Machine Learning Repository.

Dataset size: 10,000 records and 14 CSV columns. The main predictive inputs are Type, air temperature, process temperature, rotational speed, torque and tool wear. The target is Machine failure (0/1).

Expected task: Binary classification.

Suitability: The dataset supports meaningful EDA, categorical encoding, scaling, outlier investigation, class-imbalance analysis, feature engineering, leakage prevention, comparison of at least four ML models, hyperparameter tuning and an end-to-end prediction system.

Data-quality observations: The uploaded CSV contains no missing values and no exact duplicate rows. The target is highly imbalanced (339 failure records, 3.39%). Identifier columns should be excluded, and failure-mode flags must not be used as predictors of the overall Machine failure target because of leakage risk.

Ethical/licensing/accessibility considerations: The dataset is publicly accessible and licensed CC BY 4.0. It is synthetic, which reduces privacy concerns but limits claims about real industrial deployment.

23. Immediate Next Steps

1. Submit the dataset proposal and obtain instructor validation before starting formal EDA/model training.
2. After validation, create a reproducible notebook/script for EDA and preprocessing.
3. Freeze the leakage-safe predictor set and target definition.
4. Create a stratified train/test split and baseline models.
5. Assign primary model ownership to the four group members while keeping shared documentation.
6. Tune and compare models using the same evaluation framework.
7. Save the selected preprocessing pipeline and final model.
8. Build the backend/frontend prototype and test it end to end.
9. Prepare individual viva notes showing each member's contribution and understanding of the entire workflow.

24. References

- UCI Machine Learning Repository. AI4I 2020 Predictive Maintenance Dataset. DOI: 10.24432/C5HS5C. License: CC BY 4.0.
- Matzka, S. (2020). Explainable Artificial Intelligence for Predictive Maintenance Applications. International Conference on Artificial Intelligence for Industries.
- IT3051 - Fundamentals of Data Mining, Mini Project Assignment 2026. Project workflow and technical requirements.

Prepared as a project-planning document. Formal project development should begin only after dataset validation by the instructor.